

All 'Significant' Findings Are Not Created Equal

A new framework for stratifying evidence quality beyond the p-value.



Based on the study: Heston TF (2025). Significance, Fragility, and Robustness in Clinical Trials: Stratifying Statistical Evidence. *Cureus* 17(12): e100494. DOI: 10.7759/cureus.100494.

Cureus

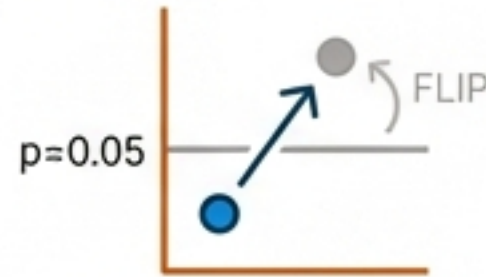
The P-Value's Blind Spot: One Answer That Hides Two Critical Questions

The reproducibility crisis has exposed limitations in our reliance on p-values. The binary classification of “significant” ($p \leq 0.05$) vs. “not significant” ($p > 0.05$) provides only partial evidence.



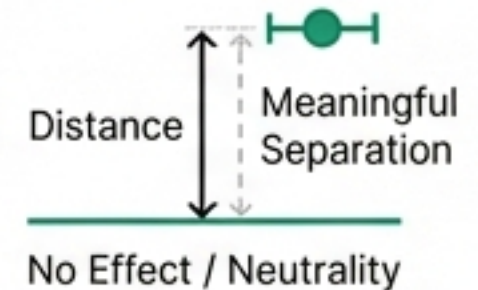
1. Classification Stability

How stable is the conclusion?
How easily does the result flip from significant to non-significant?



2. Distance from Neutrality

How far is the finding from a true “no effect”? How meaningful is the separation from therapeutic neutrality?

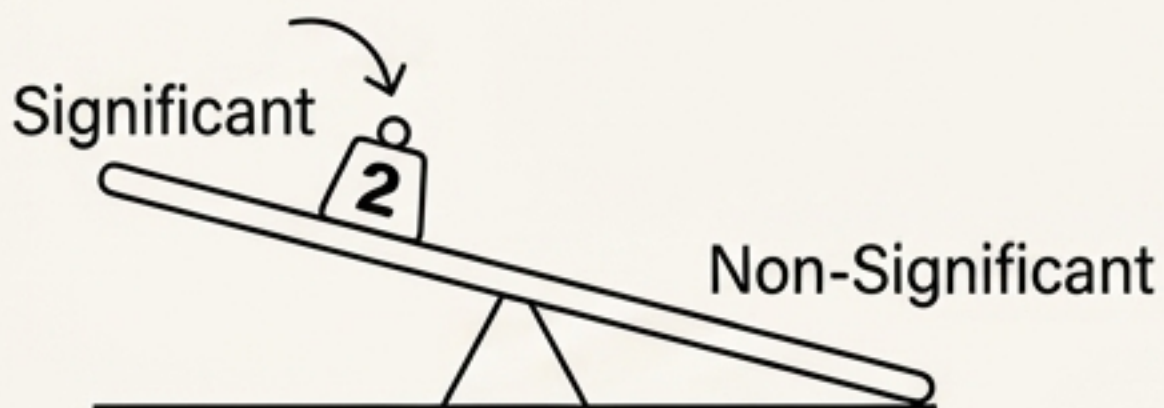


P-values address the probability of observing data under the null hypothesis, but they don't answer these crucial questions about the *quality* and *stability* of the evidence.

Consider Two Trials. The P-Value Calls Them Identical.

Trial A

P-Value: **0.04**



Flipping just **2** patient outcomes
reverses significance.

FRAGILE

Trial B

P-Value: **0.04**



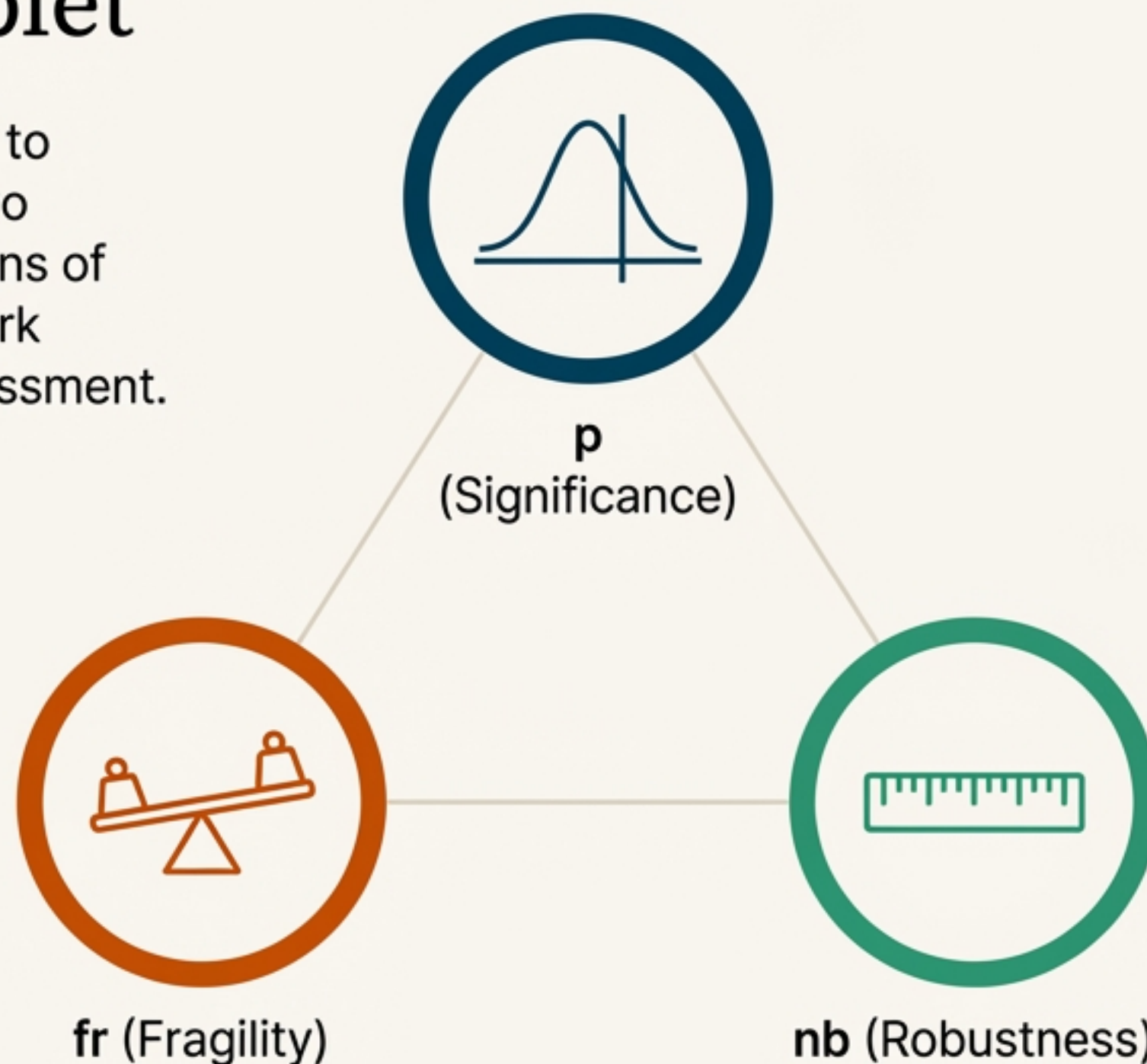
Requires **10** patient outcome
changes to reverse significance.

STABLE

Both trials are published as “statistically significant,” yet their underlying stability is vastly different. This hidden vulnerability is what we need to measure.

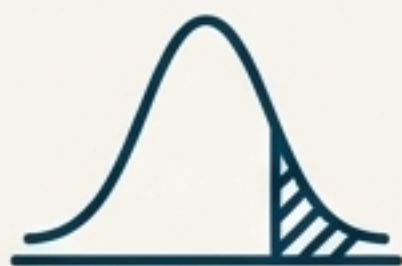
A Framework for Clarity: The p-fr-nb Triplet

To see the full picture, we need to supplement the p-value with two additional, orthogonal dimensions of evidence. The p-fr-nb framework provides a more complete assessment.



This triplet assesses completeness across three statistical inferential dimensions; it stratifies evidence quality but does not prove truth, causality, or replication.

The Three Dimensions of Statistical Evidence



Significance (p)

What is the compatibility with the null hypothesis?

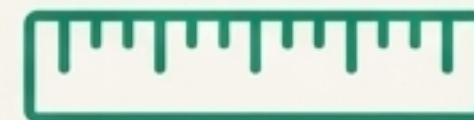
The probability of observing data at least as extreme as those obtained if no treatment effect exists. This is what we already use.



Fragility (fr)

How stable is the significance classification?

Quantifies classification stability: the minimum change in outcomes required to flip the significance decision across the $p = 0.05$ threshold. Measured by the Modified-Arm Fragility Quotient (MFQ).



Robustness (nb)

How far is the result from “no effect”?

Quantifies the geometric distance from the neutrality boundary on a standardized 0-1 scale. Measured by the Risk Quotient (RQ).

Identifying Patterns of Evidence Quality

The framework allows us to identify and classify distinct patterns of evidence that are invisible when using the p-value alone.

Concordant-Positive (CP): The Gold Standard

The simultaneous occurrence of statistical significance, high stability, and strong robustness.

IBM Plex Mono

$p \leq 0.05$, $MFQ > 0.10$, $RQ \geq 0.227$

Represents convincing evidence.

Significant-Fragile-Weak (SFW): The Precarious Boundary

Results that achieve statistical significance yet are easy to overturn and lie close to the 'no effect' boundary.

IBM Plex Mono

$p \leq 0.05$, $MFQ \leq 0.10$, $RQ < 0.075$

Represents boundary-zone evidence that warrants caution.

Putting the Framework to the Test: A Two-Pronged Validation



Monte Carlo Simulation

Generated 720,000 random trials (including 120,000 null trials where $RR=1.0$) to model a universe without selection or bias.

Purpose: To establish a “null expectation.” How often does strong evidence (CP) appear by chance alone?



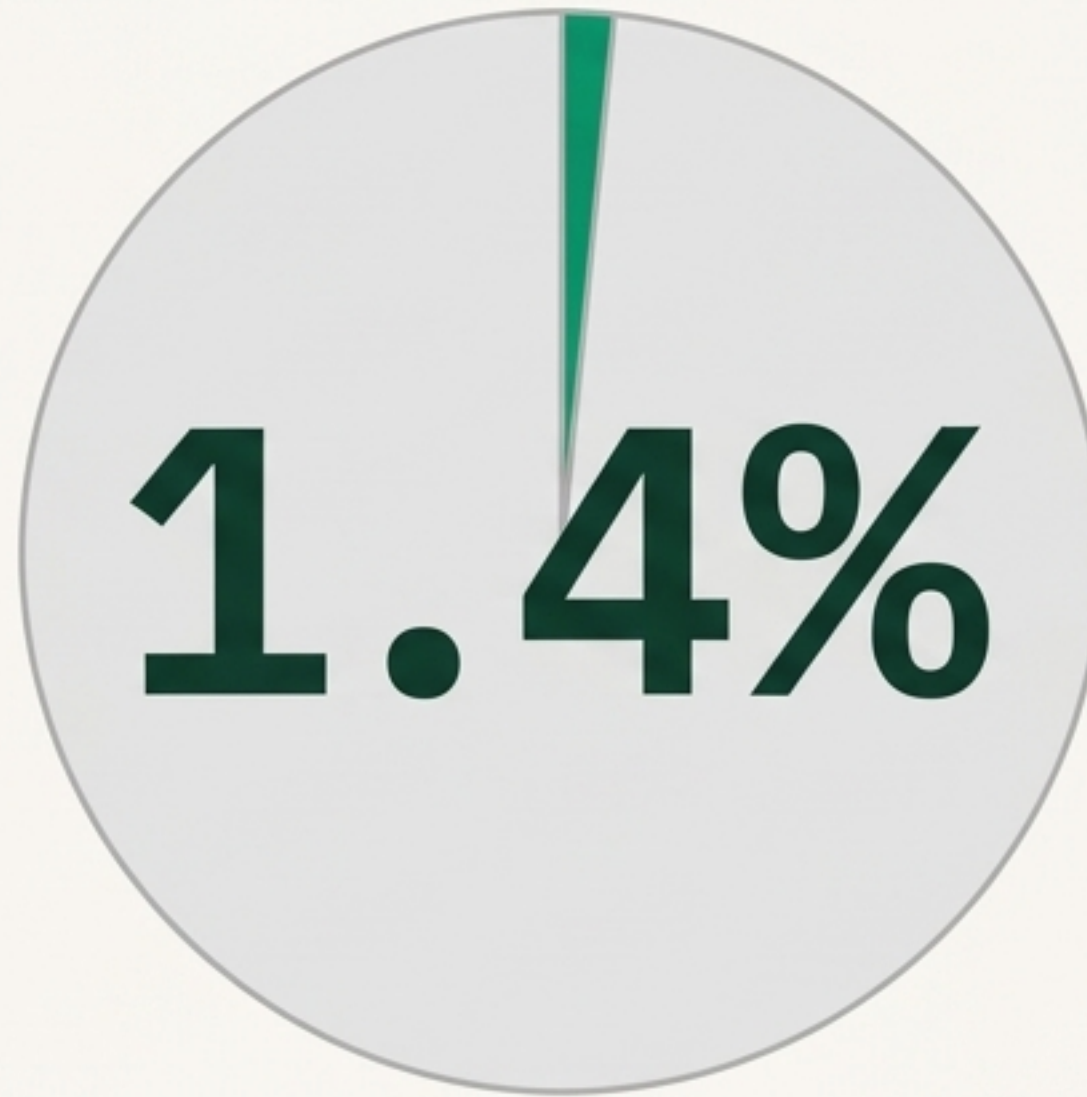
Empirical Analysis

Analyzed a pragmatic sample of 129 published, two-arm, binary-outcome clinical trials across 15 specialties.

Purpose: To measure reality. What is the actual prevalence of evidence patterns in the published literature?

The Baseline: Under the Null Hypothesis, Convincing Evidence is Exceedingly Rare

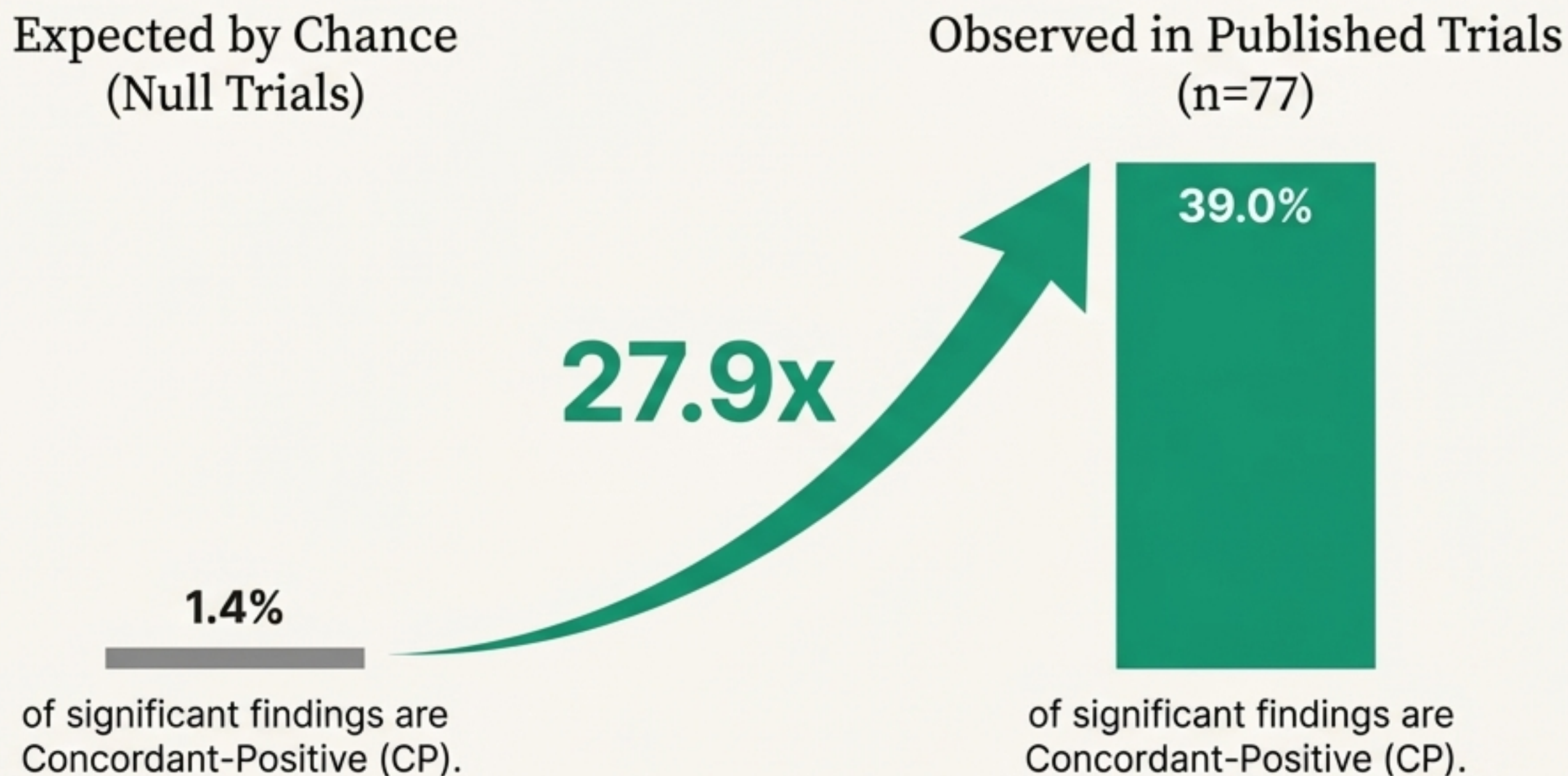
In our simulation of 120,000 trials with no true effect (Relative Risk = 1.0), 4.0% were significant ($p \leq 0.05$), as expected. We then applied the framework to these “false positives.”



Of all significant findings generated under the null hypothesis, only **1.4%** met the Concordant-Positive (CP) criteria for high-quality evidence.

This establishes a critical benchmark: strong, stable, and robust findings are not a common feature of random noise.

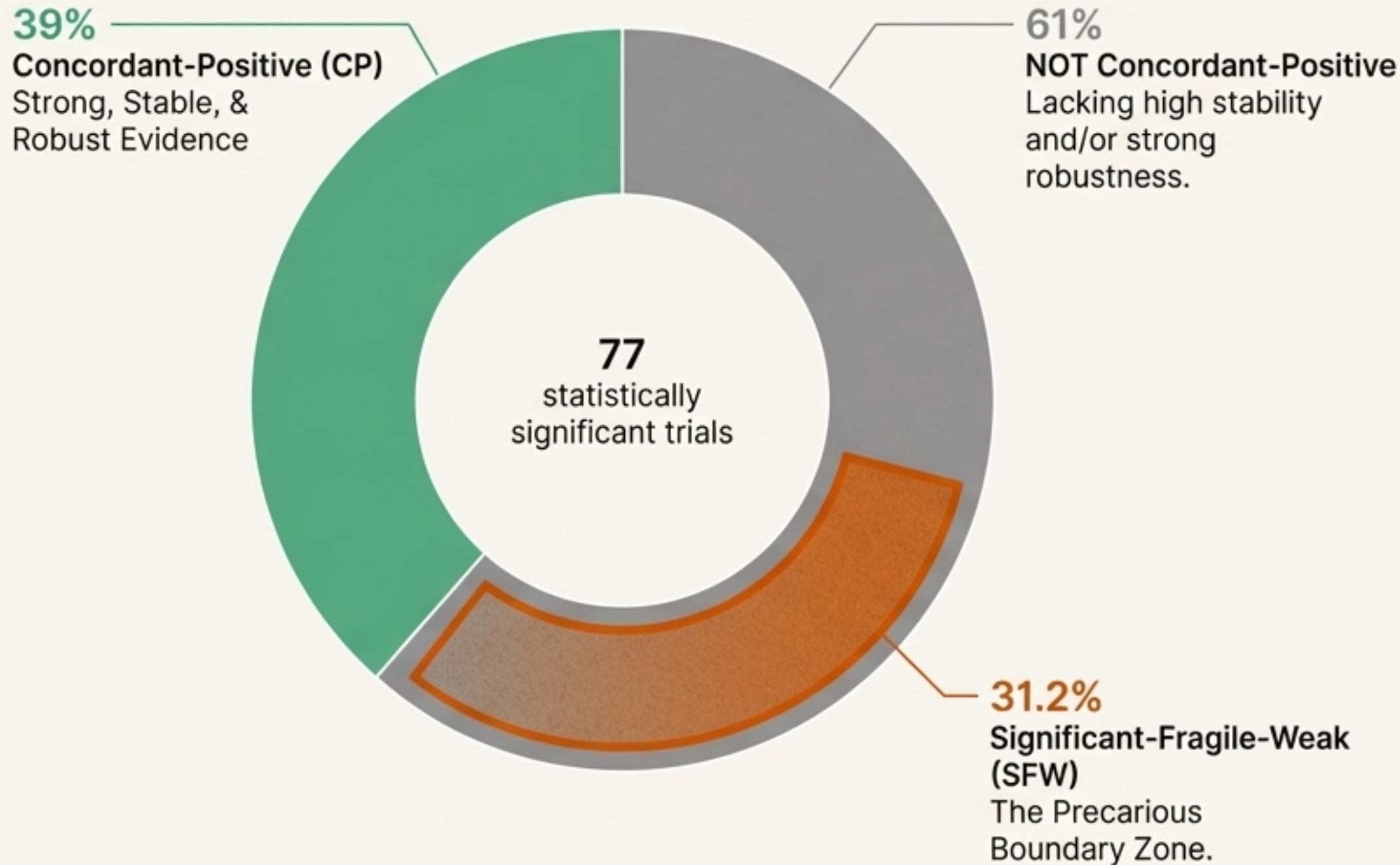
The Reveal: Published Trials Are Enriched for High-Quality Evidence



Published trials show a **27.9-fold** enrichment for high-quality evidence compared to the null expectation ($p < 0.0001$). This reflects the cumulative effect of trial design, peer review, and publication standards.

But “Significant” Is Not the Whole Story: P-Values Mask Alarming Vulnerabilities

P-Values Mask Alarming Vulnerabilities



Key Findings



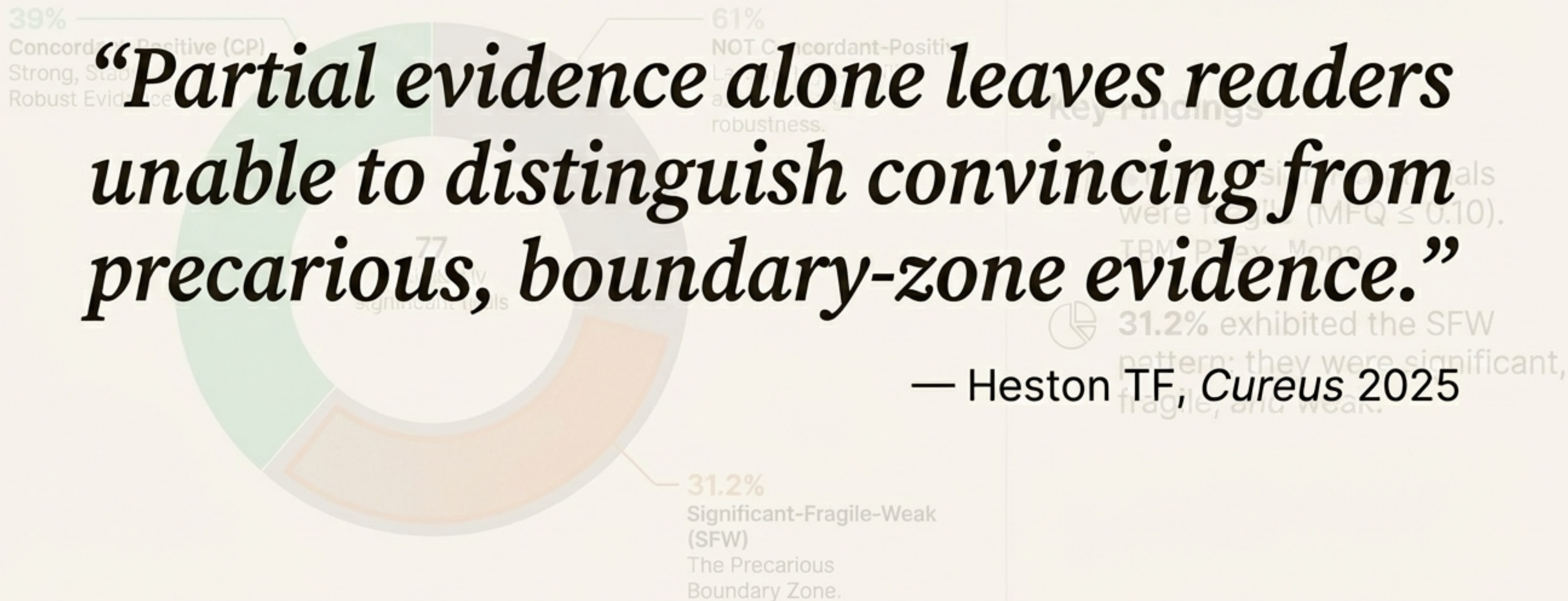
61.0% of significant trials were fragile ($\text{MFQ} \leq 0.10$).
IBM Plex Mono



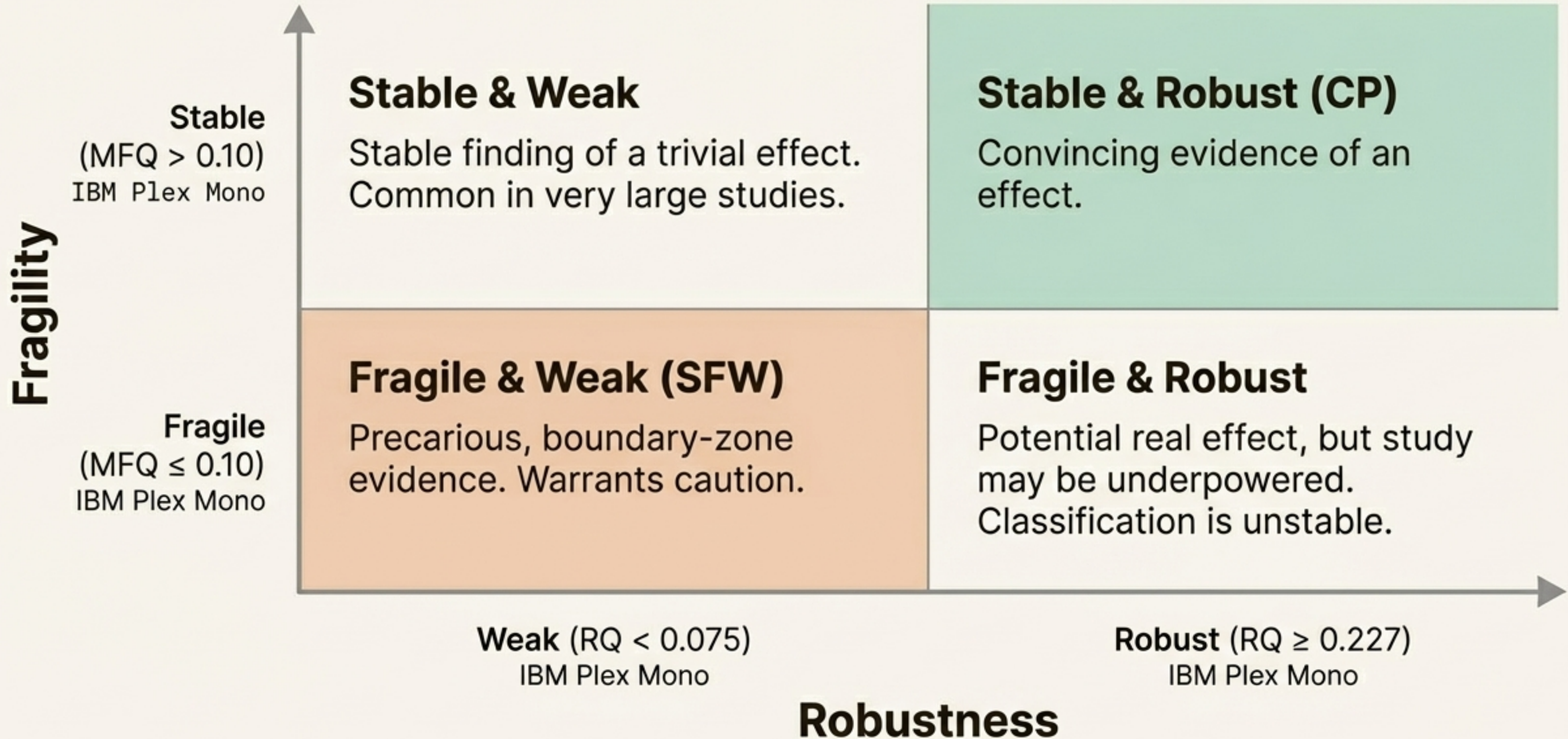
31.2% exhibited the SFW pattern: they were significant, fragile, *and* weak.

But “Significant” Is Not the Whole Story: P-Values Mask Alarming Vulnerabilities

P-Values Mask Alarming Vulnerabilities



A New Lens for Evidence: Four Quadrants of a Finding



Enhancing Rigor Across the Research Ecosystem

Adopting the p-fr-nb framework provides tangible benefits for key stakeholders, moving us toward a more reproducible and trustworthy science.



Researchers & Trialists

Go beyond a simple p-value to demonstrate the high quality and stability of your findings. Distinguish your work from more precarious results.



Journal Editors & Peer Reviewers

More effectively triage manuscripts and identify significant findings that warrant closer scrutiny due to fragility or weak robustness.



Clinicians & Guideline Authors

Better assess the true strength of the evidence behind clinical recommendations, distinguishing robust findings from those on a statistical knife's edge.

It's Time to Evolve Beyond ' $p \leq 0.05$ '

The p-fr-nb framework offers a practical, implementable solution for assessing evidence quality. Fragility and robustness metrics provide interpretive dimensions not captured by p-values and confidence intervals alone.

Recommendation

We support the evaluation and incorporation of **fragility (fr)** and **robustness (nb)** metrics into clinical trial reporting standards, such as the **CONSORT guidelines**.

The incremental reporting burden is minimal—two additional numbers—but the inferential gain is substantial.

Study Details & Resources

Full Citation

Heston TF (2025). Significance, Fragility, and Robustness in Clinical Trials: Stratifying Statistical Evidence. *Cureus* 17(12): e100494. DOI: 10.7759/cureus.100494.

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Get the Toolkit

The metrics can be computed directly from 2x2 tables using freely available software. The framework also includes extensions for continuous, ordinal, survival, and correlation analyses.

[Fragility Metrics Toolkit \(v5.2.0\) on Zenodo](#)

DOI: 10.5281/zenodo.17254763

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